

Cor ideal primo localizacion extendido

Corolario 1. *Todo ideal primo de $S^{-1}R$ es el extendido de un ideal primo de R .*

Demostración: Sea $\mathfrak{q} \subset S^{-1}R$ un ideal primo. Por el apartado (1) de la `prop-ideales-extendidos-con` tenemos que $(\mathfrak{q}^c)^c = \mathfrak{q}$. Falta ver que $\mathfrak{q}^c \subset R$ es un ideal primo.

Sean $a, b \in R$ tales que $ab \in \mathfrak{q}^c$. Entonces, $\frac{ab}{1} = \iota_S(ab) \in \mathfrak{q}$. Como ι_S es morfismo de anillos, $\frac{a}{1} \cdot \frac{b}{1} = \iota_S(a) \cdot \iota_S(b) = \iota_S(ab) \in \mathfrak{q}$. Como $\mathfrak{q}$ es primo, $\frac{a}{1} \in \mathfrak{q} \vee \frac{b}{1} \in \mathfrak{q}$, es decir, $a \in \mathfrak{q}^c \vee b \in \mathfrak{q}^c$. Por tanto, $\mathfrak{q}^c$ es primo. ■

Referenciado en

- `teo-extension-entera-ideal-primo-imp-exists-ideal-primo-contrae`
- `ejer-localizacion-ideal-primo-anillo-local`

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